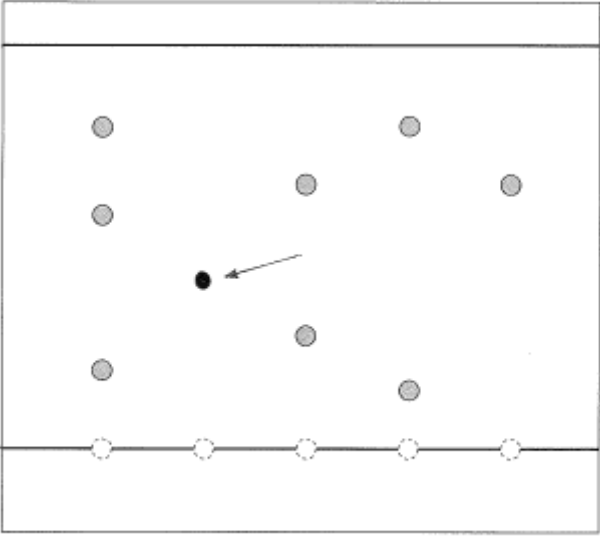


| Question |     |      |  | Marking details                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 2        | (a) |      |  | chromatography                                                                                                                                                | 1               |     |     | 1     |       | 1    |
|          | (b) | (i)  |  | award (2) for 3 / 3.0 / 3.00 / 2.996<br><br>if incorrect award (1) for correct distance moved by solvent front i.e. 7 cm                                      |                 | 2   |     | 2     | 2     | 2    |
|          |     | (ii) |  |  <p>spot added 3 cm above start line</p> <p>ecf possible from part (i)</p> |                 |     | 1   | 1     |       |      |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                | Marks available |          |          |          |          |          |
|----------|-----|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
|          | (c) |  |  | <b>C</b> (1)<br><br>any of following for (1) <ul style="list-style-type: none"><li>• spot corresponds with that of banned substance</li><li>• spot at 4.6 cm as it is for banned substance</li><li>• dye has the same <math>R_f</math> value as the banned substance</li></ul> |                 |          | 2        | 2        |          | 2        |
|          |     |  |  | <b>Question 2 total</b>                                                                                                                                                                                                                                                        | <b>1</b>        | <b>2</b> | <b>3</b> | <b>6</b> | <b>2</b> | <b>5</b> |